

Quiz — 1.4.3 Boolean Algebra — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (8 marks — 1 each) 1. **D** — AND outputs 1 only when both inputs are 1. 2. **C** — XOR outputs 1 when inputs differ. 3. **B** — NAND and NOR are the universal gates. 4. **B** — $\neg(A + B) = \neg A \cdot \neg B$ (break the bar, swap OR $\rightarrow$ AND). 5. **C** — double negation. 6. **C** — half adder Sum = $A \oplus B$. 7. **B** — D-type flip-flop stores one bit on a clock edge. 8. **C** — K-map groups are powers of two and as large as possible.

Section B

9. [5] $Q = \neg(A \cdot B) + (A \cdot C)$:

A	B	C	A·B	$\neg(A \cdot B)$	A·C	Q
0	0	0	0	1	0	1
0	0	1	0	1	0	1
0	1	0	0	1	0	1
0	1	1	0	1	0	1
1	0	0	0	1	0	1
1	0	1	0	1	1	1
1	1	0	1	0	0	0
1	1	1	1	0	1	1

Mark: A·B column correct (1); $\neg(A \cdot B)$ column correct (1); A·C column correct (1); Q correct for rows 1–4 (1); Q correct for rows 5–8 — note only row A=1, B=1, C=0 gives Q=0 (1).

10. [3] Simplify $\neg(\neg A + B)$: - Apply De Morgan's to the outer bar: $\neg(\neg A + B) = \neg(\neg A) \cdot \neg B$ (break the bar, change + to ·, negate each term). (1) - Apply double negation: $\neg(\neg A) = A$. (1) - Result: $A \cdot \neg B$. (1)

11. [3] Half adder with A = 1, B = 1: - Sum = $A \oplus B = 1 \oplus 1 = 0$ (1) - Carry = $A \cdot B = 1 \cdot 1 = 1$ (1) - Correct expressions stated (Sum = $A \oplus B$, Carry = $A \cdot B$) (1). (So $1 + 1 =$ binary 10 : Sum 0, Carry 1.)

12. [3] Half adder truth table:

A	B	Sum ($A \oplus B$)	Carry ($A \cdot B$)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Mark: Sum column correct (1); Carry column correct (1); fully correct table with 1+1 giving Sum 0 Carry 1 (1).

13. [3] $A \cdot (A + B)$: - This is the **Absorption law**: $A \cdot (A + B) = A$. (1 for naming Absorption) - Result: **A**. (1) - Justification (1): expand $A \cdot (A+B) = A \cdot A + A \cdot B = A + A \cdot B$ (idempotent $A \cdot A = A$); then $A + A \cdot B = A$ (A absorbs $A \cdot B$), so the whole expression reduces to A.

Section C

14. [5] (a) **[1]** $\text{Sum} = A \oplus B \oplus \text{Cin}$.

(b) **[2]** $\text{Cout} = (A \cdot B) + (\text{Cin} \cdot (A \oplus B))$. (1 mark for the $A \cdot B$ term, 1 mark for the $\text{Cin} \cdot (A \oplus B)$ term ORed together. Equivalent correct forms such as $A \cdot B + B \cdot \text{Cin} + A \cdot \text{Cin}$ also gain full marks.)

(c) **[2]** A **half adder** adds only **two** bits (A and B) and has **no carry-in**, so it cannot accept a carry from a previous column (1). A **full adder** adds **three** bits (A, B and Cin), so full adders can be **chained** (a ripple-carry adder) with each carry-out feeding the next column's carry-in, allowing multi-bit binary numbers to be added correctly (1).

Total: $8 + 17 + 5 = 30$ marks.